SRINIVAS UNIVERSITY

INSTITUTE OF ENGINEERING & TECHNOLOGY

MUKKA, MANGALURU-574146

MINI PROJECT REPORT

ON

SMART HEALTHCARE APPOINTMENT SYSTEM

Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN

CYBER SECURITY

Submitted By,

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DEPARTMENT OF CYBER SECURITY

SRINIVAS UNIVERSITY, MUKKA

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CERTIFICATE

This is to certify that the Technical Seminar entitled ***"Smart Healthcare Appointment System"*** is a bonafide work carried out by ***Anagha Arun, Sanjana Anil Naik*** bearing the **USN** ***01SU23CF002 , 01SU23CB024***in the partial fulfillment for the award of **Bachelor of Technology** in **Cyber Security** of the **Srinivas University-Institute of Engineering & Technology** during the year **2025-26.** It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The mini project report has been approved as it meets the academic criteria outlined for the said degree.

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DECLARATION

We, ***Anagha Arun ,Sanjana Anil Naik*** the student of 6th semester, **Bachelor of Technology** in **Cyber Security** of the **Srinivas University- Institute of Engineering & Technology**, Mukka, hereby declare that the mini project report entitled **"Smart Healthcare Appointment System"** has been successfully completed by us in partial fulfillment of the requirements for the award of degree in **Bachelor of Technology** in **Cyber Security of Srinivas University- Institute of Engineering & Technology** and no part of it has been submitted for the award of degree or diploma in any university or institution previously.

Date: 25-03-2026
Place: Mukka

ABSTRACT

This project presents the design and implementation of a **Microservices-Based Smart Healthcare Appointment System** that demonstrates modern software architecture principles. The system is developed using a **microservices architecture pattern**, where the application is divided into multiple independent services that work together to provide healthcare appointment management. Each service is responsible for a specific functionality and communicates with other services through well-defined APIs.

The healthcare appointment system consists of several microservices such as **API Gateway, User Service, Doctor/Appointment Service, and Notification Service**. Each microservice operates independently and manages its own data, ensuring better scalability, flexibility, and maintainability. The system allows patients to register, log in, view available doctors, and book appointments online. It also enables efficient management of appointment records and user information.

The frontend of the application is developed using **React**, providing a responsive and user-friendly interface for patients. The backend services are implemented using **Node.js and Express.js**, while **MongoDB** is used as the database to store patient details, doctor information, and appointment data. The services communicate through APIs via the API Gateway, which acts as the single entry point for client requests.

The entire system can be containerized using **Docker**, which simplifies deployment and improves scalability in distributed environments. This microservices-based approach enhances system reliability, allows independent development of services, and improves overall performance compared to traditional monolithic systems.

Keywords: Microservices Architecture, Smart Healthcare System, Node.js, React, MongoDB, API Gateway, Online Appointment System.

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Chapter-1

INTRODUCTION

Healthcare services are an essential part of modern society, and efficient management of patient appointments is important for providing quality medical care. In many hospitals and clinics, appointment scheduling is still handled manually, which can lead to long waiting times, overcrowding, and inefficient management of patient records. To overcome these issues, digital healthcare solutions have become increasingly important.

The **Smart Healthcare Appointment System** is a web-based application designed to simplify the process of booking and managing medical appointments. The system allows patients to easily register, log in, view available doctors, and book appointments through an online platform. This reduces the need for physical visits to hospitals just to schedule appointments and improves the overall efficiency of healthcare services.

This project is developed using a **microservices architecture**, where the application is divided into multiple independent services. Each service performs a specific function such as user management, appointment handling, or notifications. These services communicate with each other through **Application Programming Interfaces (APIs)**. This approach makes the system more scalable, flexible, and easier to maintain compared to traditional monolithic systems.

The system includes several main components such as **API Gateway, User Service, Doctor/Appointment Service, and Notification Service**. The API Gateway acts as the main entry point for all client requests and routes them to the appropriate service. Each microservice operates independently and can be developed or updated without affecting the entire system.

The frontend of the system is developed using **React**, which provides a responsive and interactive user interface. The backend services are implemented using **Node.js and Express.js**, while **MongoDB** is used as the database for storing patient information, doctor details, and appointment records. These technologies together create a modern and efficient healthcare appointment management system.

The Smart Healthcare Appointment System helps improve hospital management by reducing manual work, minimizing waiting time, and providing a convenient way for patients to schedule medical consultations online.

Chapter 2

PROBLEM STATEMENT

In many hospitals and healthcare centers, the process of booking appointments is still performed manually or through basic systems that are not efficient enough to handle large numbers of patients. Patients often need to visit the hospital in person or make phone calls to schedule appointments with doctors. This traditional method can lead to several problems such as long waiting times, overcrowding, poor management of appointment records, and lack of proper communication between patients and healthcare providers.

Manual appointment systems also make it difficult for hospitals to manage patient information, doctor schedules, and appointment history efficiently. As the number of patients increases, maintaining accurate records and coordinating appointments becomes more complex and time-consuming. In addition, traditional monolithic software systems used in some hospitals are not flexible enough to scale or adapt to new requirements.

With the advancement of modern web technologies, there is a need for a **smart and scalable healthcare appointment system** that allows patients to easily book appointments online and enables hospitals to manage appointment data efficiently.

Therefore, this project proposes the development of a **Microservices-Based Smart Healthcare Appointment System** that provides an online platform for patient registration, doctor viewing, and appointment booking. By using microservices architecture, the system divides different functionalities into independent services such as user management, appointment handling, and notifications. This improves system scalability, flexibility, and maintainability.

The proposed system aims to reduce manual effort, minimize waiting times for patients, and provide a more efficient and organized method for managing healthcare appointments.

Chapter 3

OBJECTIVES OF THE PROJECT

The main objective of the Smart Healthcare Appointment System is to develop an efficient, reliable, and user-friendly platform that enables patients to book and manage medical appointments online. The system aims to enhance healthcare service management by reducing manual effort, minimizing waiting time, and ensuring a more organized and streamlined appointment scheduling process. By replacing traditional methods with a digital solution, the system improves accessibility and convenience for both patients and healthcare providers.

The project focuses on implementing a microservices-based architecture in which core functionalities such as user management, appointment handling, and notifications are separated into independent services. This approach improves system scalability, flexibility, and ease of maintenance, allowing each service to be developed and updated independently. The system also provides a responsive and intuitive user interface, enabling patients to easily navigate, register, log in, view doctor details, and book appointments without difficulty.

Another key objective is to ensure secure user authentication and efficient management of patient and doctor data. The system uses MongoDB for flexible and scalable data storage, which allows efficient handling of large amounts of information such as user profiles, doctor details, and appointment records. In addition, an API Gateway is implemented to act as a central communication point between the frontend and backend services, ensuring smooth and organized data flow across the system.

Furthermore, the project aims to improve overall healthcare service quality by enhancing communication between patients and the system through notification services. These services provide appointment confirmations and updates, helping users stay informed about their bookings. The system is also designed to be scalable so that it can handle increasing numbers of users and services in the future without affecting performance.

Overall, the goal of the project is to deliver a smart, scalable, and efficient healthcare solution that improves patient convenience, optimizes hospital appointment management, and demonstrates the effective use of modern technologies and microservices architecture in real-world applications.

Chapter-4

SYSTEM ARCHITECTURE

The **Smart Healthcare Appointment System** is designed using a **microservices architecture**, where the application is divided into multiple independent services. Each service performs a specific function and communicates with other services through APIs. This approach improves scalability, flexibility, and maintainability compared to traditional monolithic systems.
The system architecture consists of the following main components:

4.1 Frontend

The frontend is the user interface of the system, developed using **React**. It allows patients to interact with the application easily.
Main features of the frontend include:

- User registration and login
- Viewing available doctors
- Booking appointments
- Viewing appointment details

The frontend communicates with backend services through the API Gateway.

4.2 API Gateway

The **API Gateway** acts as the central entry point for all client requests. Instead of directly interacting with individual services, the frontend sends requests to the API Gateway.
Functions of the API Gateway include:

- Routing requests to appropriate microservices
- Handling client-server communication
- Simplifying system interaction

This improves security and reduces complexity in the system.

4.3 User Service

The **User Service** is responsible for managing user-related operations.
Its main functions include:

- User registration
- User authentication (login/logout)
- Managing user profile data

All user information is stored in the **MongoDB database**, ensuring secure and efficient data handling.

4.4 Doctor / Appointment Service

This service manages doctor information and appointment scheduling.
Key functions include:

- Displaying doctor details and availability
- Booking appointments
- Managing appointment records
- Updating appointment status

It ensures smooth interaction between patients and doctors.

4.5 Notification Service

The **Notification Service** handles communication with users.
Its functions include:

- Sending appointment confirmations
- Sending reminders
- Providing system notifications

This improves user experience and keeps patients informed.

4.6 Database (MongoDB)

The system uses **MongoDB**, a NoSQL database, to store all application data.
Data stored includes:

- User information
- Doctor details
- Appointment records

MongoDB's document-based structure allows flexible and scalable data management, making it suitable for microservices architecture.

4.7 Working of the System

1. The user interacts with the frontend (React interface).
2. The request is sent to the API Gateway.
3. The API Gateway routes the request to the appropriate microservice.
4. The microservice processes the request and interacts with the database if needed.
5. The response is sent back to the frontend through the API Gateway.

This architecture ensures that each service works independently, making the system easier to develop, update, and scale.
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Chapter-5

IMPLEMENTATION

The **Smart Healthcare Appointment System** is implemented using a microservices architecture where the application is divided into different modules. Each module performs a specific function and works independently while communicating with other modules through APIs. This modular approach improves system efficiency, scalability, and maintainability.
The main modules of the system are described below.

5.1 User Module

The **User Module** manages all operations related to users of the system. It allows patients to create accounts and securely access the application.
Main functions of this module include:

- User registration
- User login and authentication
- Managing user profile information

During registration, users provide details such as name, email, and password. These details are stored securely in the **MongoDB database**. After successful login, users can access other features of the system such as viewing doctors and booking appointments.

5.2 Doctor Module

The **Doctor Module** manages information related to doctors available in the healthcare system.
Main functions include:

- Displaying the list of doctors
- Showing doctor specialization and details
- Allowing users to select doctors for appointments

This module helps patients easily find and choose the appropriate doctor based on their needs.

5.3 Appointment Module

The **Appointment Module** handles the process of scheduling and managing appointments between patients and doctors.
Main functions include:

- Booking appointments
- Viewing appointment details
- Managing appointment records

When a patient selects a doctor and appointment time, the system stores the appointment details in the **MongoDB database**. This helps keep track of all scheduled consultations.

5.4 Notification Module

The **Notification Module** is responsible for sending notifications to users regarding their appointments.
Main functions include:

- Sending appointment confirmation messages
- Sending reminders for scheduled appointments
- Providing system alerts or updates

This module helps improve communication between patients and healthcare providers.

5.5 API Gateway Module

The **API Gateway Module** acts as the central communication layer between the frontend and backend services. Functions include:

- Receiving requests from the frontend
- Routing requests to the appropriate microservice
- Managing communication between different services

This module simplifies the system architecture and ensures smooth interaction between services.

These modules work together to provide a complete **Smart Healthcare Appointment System** that allows patients to manage appointments efficiently while maintaining a scalable and organized system structure.

Chapter-6

METHODOLOGY

The Smart Healthcare Appointment System is developed using a **microservices-based architecture**, which divides the system into independent services responsible for specific functionalities. This methodology ensures modularity, scalability, and easier maintenance. The system uses **React** for the frontend, **Node.js and Express.js** for backend services, and **MongoDB** as the database for storing patient, doctor, and appointment information.

The development methodology follows an **incremental and iterative approach**. The project is divided into multiple modules, and each module is designed, implemented, and tested independently. The main modules include User Management, Doctor Management, Appointment Management, Notification Service, and the API Gateway. Each module communicates with its own database collection, ensuring independent functionality and data isolation.

User Management handles patient registration, authentication, and profile management. **Doctor Management** stores and provides doctor information, including specialization and availability. **Appointment Management** enables patients to schedule, view, and cancel appointments, while the **Notification Service** sends appointment confirmations and reminders. The **API Gateway** serves as a single entry point, routing requests from the frontend to the appropriate microservice.

The methodology emphasizes **secure, efficient, and scalable system design**. MongoDB's flexible document-based structure allows for easy storage and retrieval of data, while the microservices approach ensures that updates or modifications to one module do not affect the rest of the system. The development process includes careful testing of each module to verify functionality and integration before deployment.

The overall methodology combines modern web technologies with best practices in microservices architecture to provide a reliable and user-friendly online healthcare appointment system. The system is designed to reduce manual work, minimize waiting times, and improve the overall efficiency of hospital appointment management.
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Chapter-7

DATABASE DESIGN

The Smart Healthcare Appointment System uses **MongoDB**, a NoSQL document-oriented database, for storing and managing all application data. MongoDB is chosen for its flexibility and scalability, which makes it ideal for a microservices-based architecture where each service manages its own collection independently. This allows for easy updates, high performance, and efficient handling of large volumes of user, doctor, and appointment information.

The database is organized into three primary collections: **Users**, **Doctors**, and **Appointments**. The **Users collection** stores patient details such as user ID, name, email, password, phone number, and role. This collection is managed by the User Service and is essential for user registration, login, and profile management. Passwords are stored securely using encryption to ensure privacy.

The **Doctors collection** stores information about all doctors in the system, including doctor ID, name, specialization, experience, availability status, and consultation details. The Doctor Service manages this collection, providing patients with accurate and updated doctor information for appointment booking.

The **Appointments collection** stores all appointment-related information. Each document includes appointment ID, patient ID, doctor ID, appointment date, time, and status (booked, completed, or canceled). The Appointment Service handles this collection to manage appointment scheduling, while the Notification Service uses this data to send confirmations and reminders to patients.

The database design is aligned with the microservices architecture. Each service interacts only with its respective collection, ensuring independence and reducing coupling. The document-based structure of MongoDB allows easy modification and addition of fields without affecting existing data. This provides flexibility for future enhancements, such as adding new features or data types.

Overall, the MongoDB database ensures **secure, scalable, and efficient storage** of all critical data, supporting the smooth functioning of the Smart Healthcare Appointment System while maintaining data integrity and performance.

Chapter- 8

SYSTEM FLOW AND DIAGRAMS

The Smart Healthcare Appointment System is designed to provide patients with an easy-to-use platform for booking and managing appointments with doctors. The system follows a **microservices-based architecture**, where each service operates independently, performing a specific function. The **API Gateway** acts as a central hub for all communication between the frontend and the backend services. MongoDB is used as the database to store all relevant information securely.

9.1 System Flow Overview

The system flow can be divided into **five main stages**:

1. **User Interaction:**

   The patient accesses the system through a **React-based frontend**, which provides an interactive and responsive interface. The patient can perform the following actions:
   1. Register as a new user
   2. Log in to their account
   3. View the list of available doctors
   4. Book, cancel, or view appointments
   5. Update their profile information

2. **API Gateway Routing:**

   All requests from the frontend are sent to the **API Gateway**. This acts as the entry point for all client requests and routes them to the appropriate backend service. For example:
   1. User registration requests go to the **User Service**
   2. Appointment booking requests go to the **Appointment Service**
   3. Doctor information requests go to the **Doctor Service**

3. **Microservice Processing:**

   Each microservice handles a specific functionality:
   1. **User Service:** Handles registration, authentication, and profile management. Interacts with the **Users collection** in MongoDB.
   2. **Doctor Service:** Provides doctor information, including specialization, experience, and availability. Retrieves data from the **Doctors collection** in MongoDB.
   3. **Appointment Service:** Manages appointment creation, updates, and cancellation. Stores and retrieves data from the **Appointments collection** in MongoDB.
   4. **Notification Service:** Reads data from the **Appointments collection** to send confirmations and reminders via email or SMS.

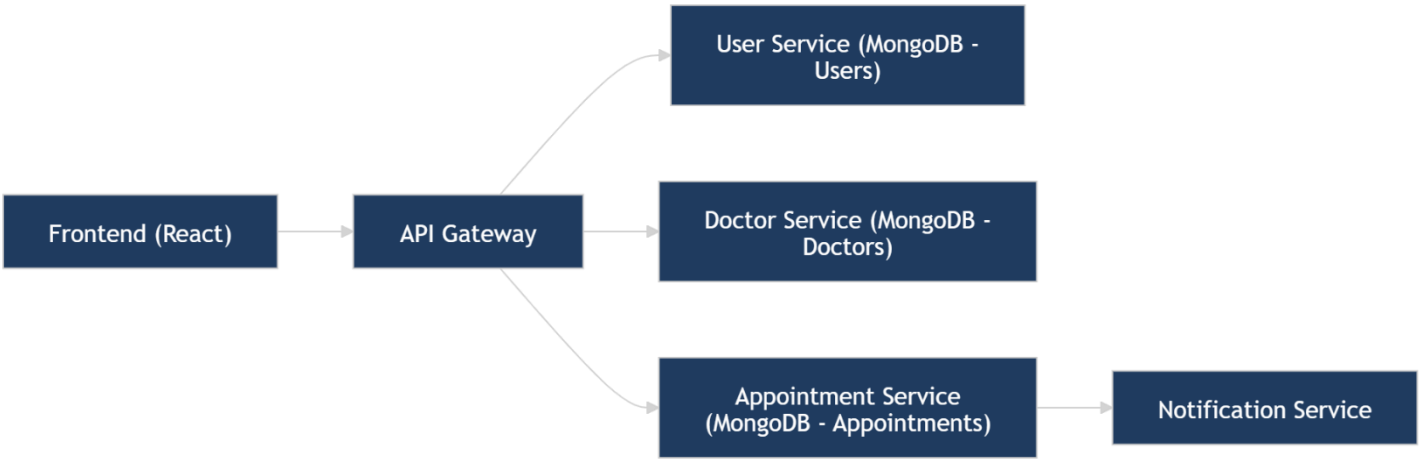
4. **Database Operations:**

   Each microservice interacts only with its corresponding database collection. MongoDB's flexible, document-based structure allows each collection to store detailed information and ensures that updates or queries can be performed efficiently without affecting other services.

5. **Response Back to Frontend:**
   After processing, the microservices send responses back to the **API Gateway**, which forwards the data to the frontend. The patient then receives a confirmation of actions such as a booked appointment, updated profile, or a list of doctors.

9.2 System Flow Diagram

Here's a **high-level overview** of the system flow:



Explanation:

- The **Frontend** is where the user interacts with the system.
- The **API Gateway** directs requests to the correct microservice.
- **Microservices** handle processing and communicate with MongoDB.
- The **Notification Service** ensures that users receive updates about their appointments.

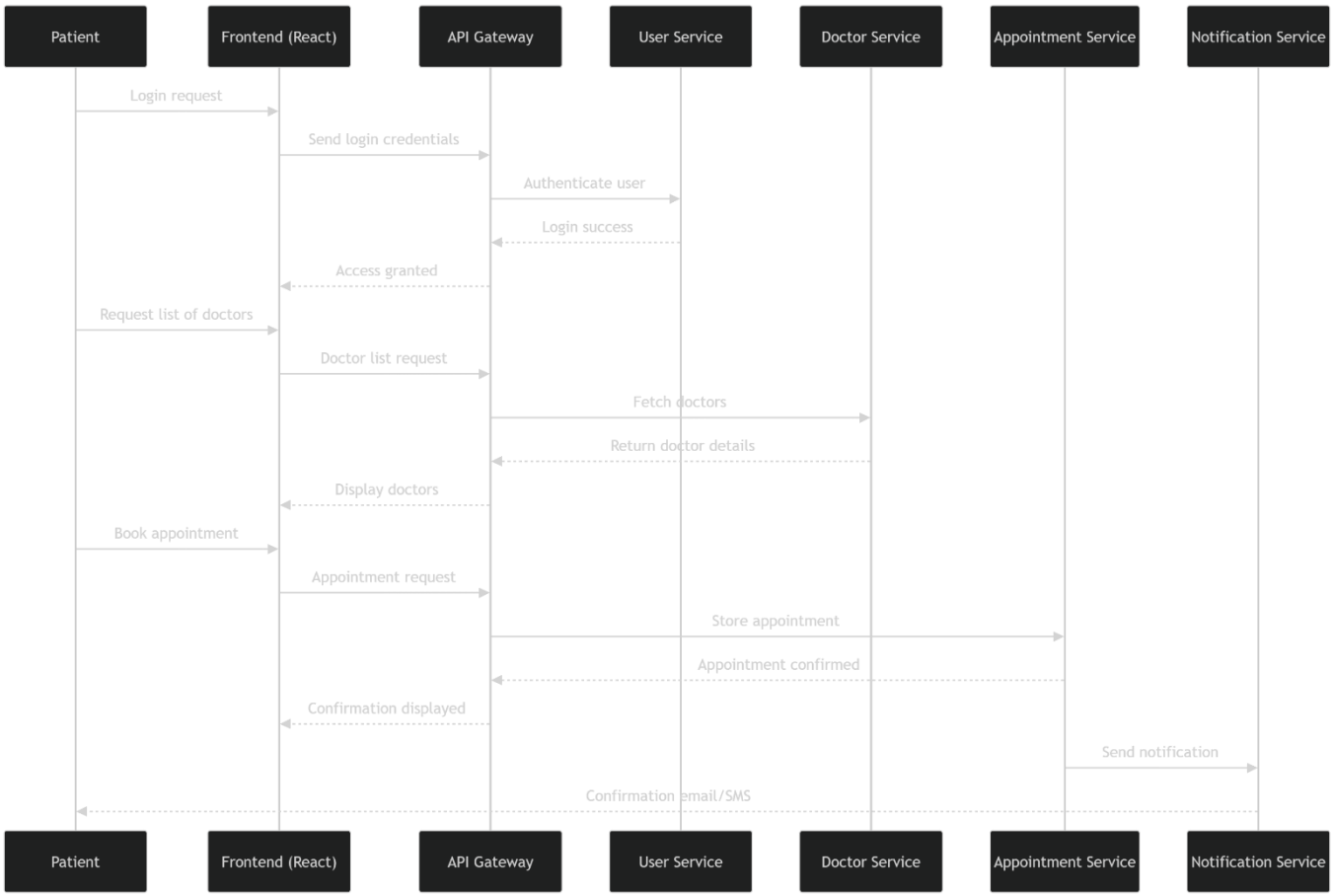
9.3 Detailed Flow of Booking an Appointment

The **appointment booking process** involves several steps, which can be visualized as a sequence:

1. The patient logs in via the frontend.
2. The frontend sends a request to the **API Gateway** for authentication.
3. The API Gateway forwards the request to the **User Service**.
4. The User Service validates the login credentials using the **Users collection** in MongoDB.
5. Upon successful login, the frontend requests a list of available doctors.
6. The API Gateway routes this request to the **Doctor Service**, which retrieves doctor details from the **Doctors collection**.

7. The patient selects a doctor and appointment time.
8. The API Gateway sends the booking request to the **Appointment Service**, which stores the appointment in the **Appointments collection**.
9. The **Notification Service** sends a confirmation to the patient's email or phone.
10. The frontend displays a confirmation message to the patient.

9.4 Sequence Diagram for Appointment Booking



Explanation:
This sequence diagram shows step-by-step how a user books an appointment and how the system processes the request across services and database collections.

Chapter-9

RESULT

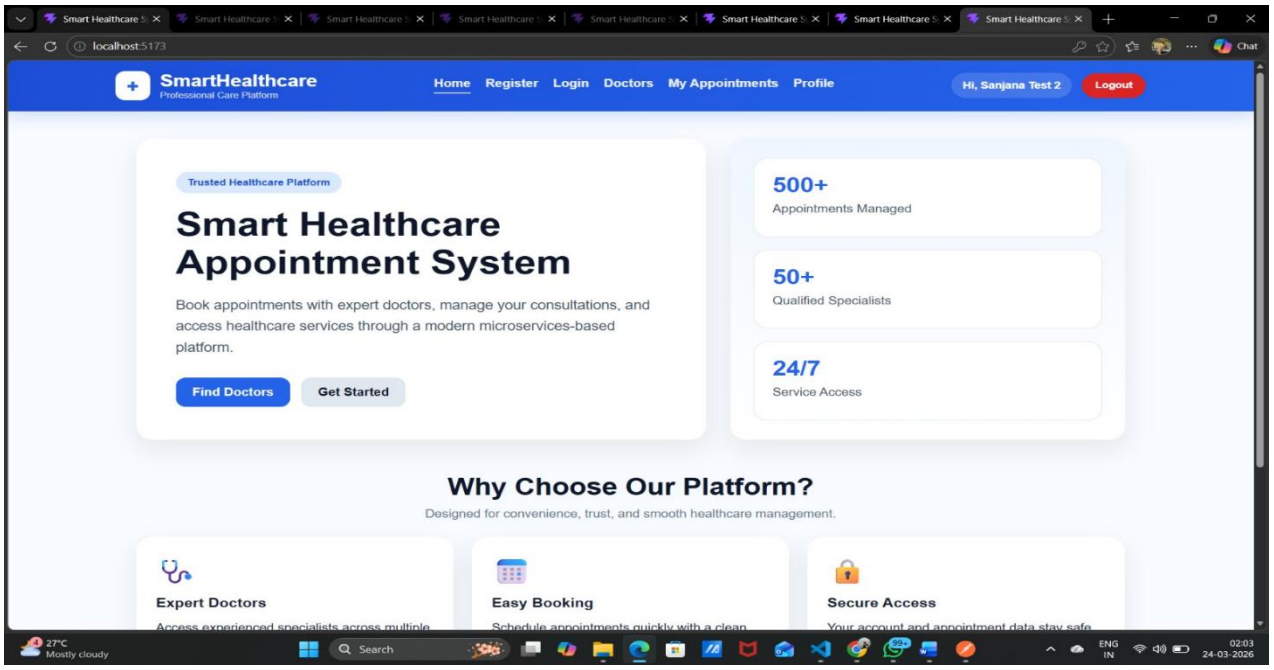


Fig:9.1 User Dashboard Interface

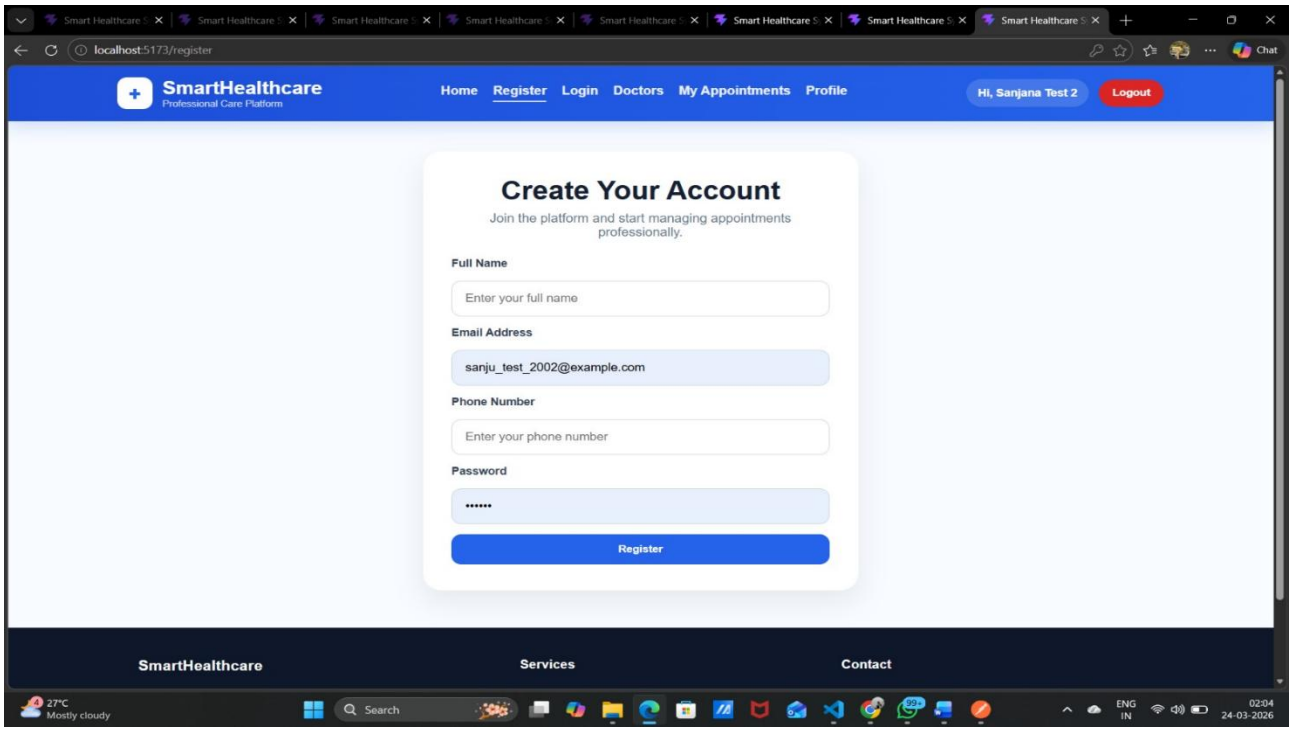


Fig 9.2 Register panel

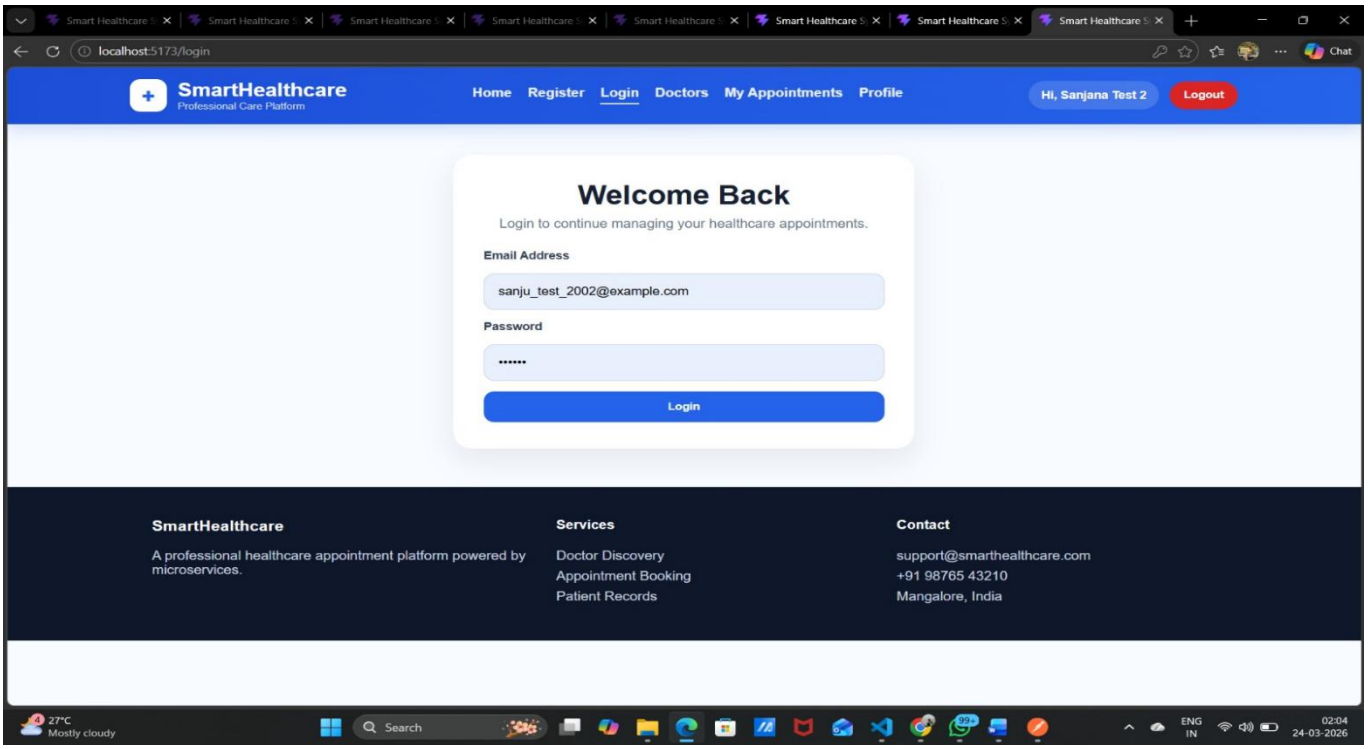


Fig 9.3 Login panel

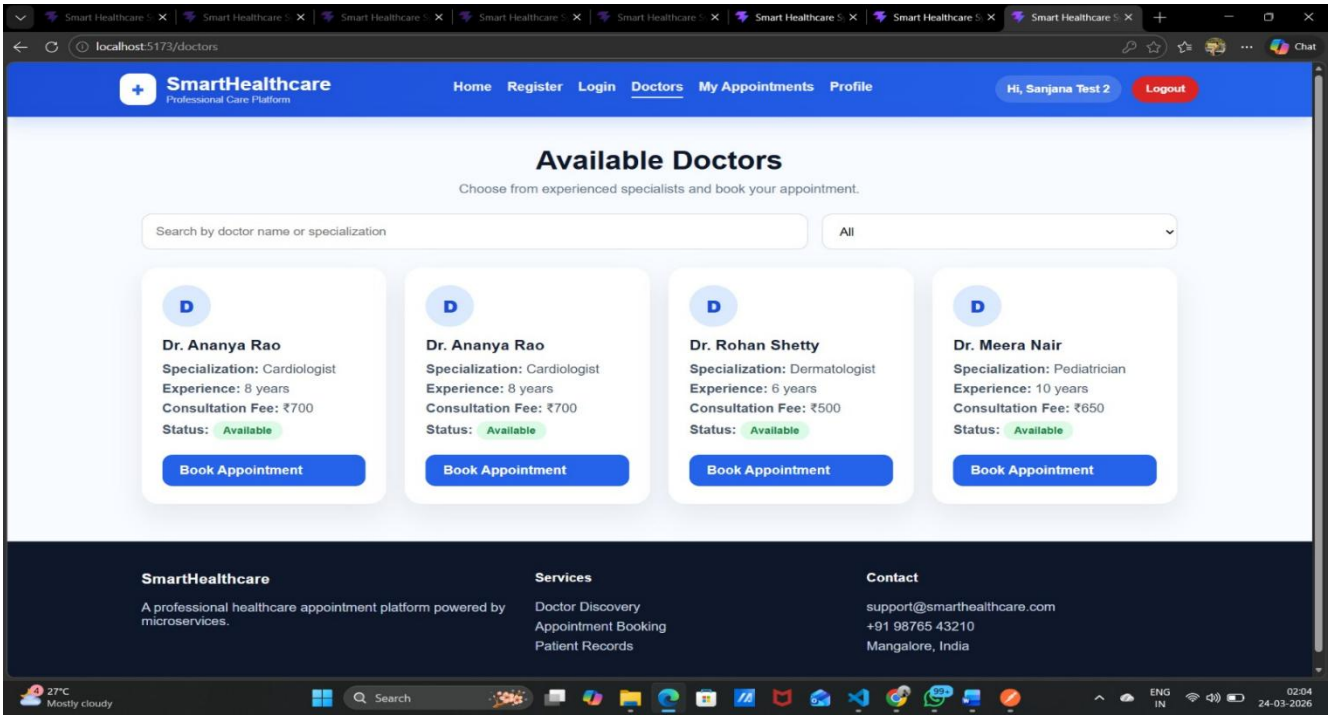


Fig 9.4 Available Doctor

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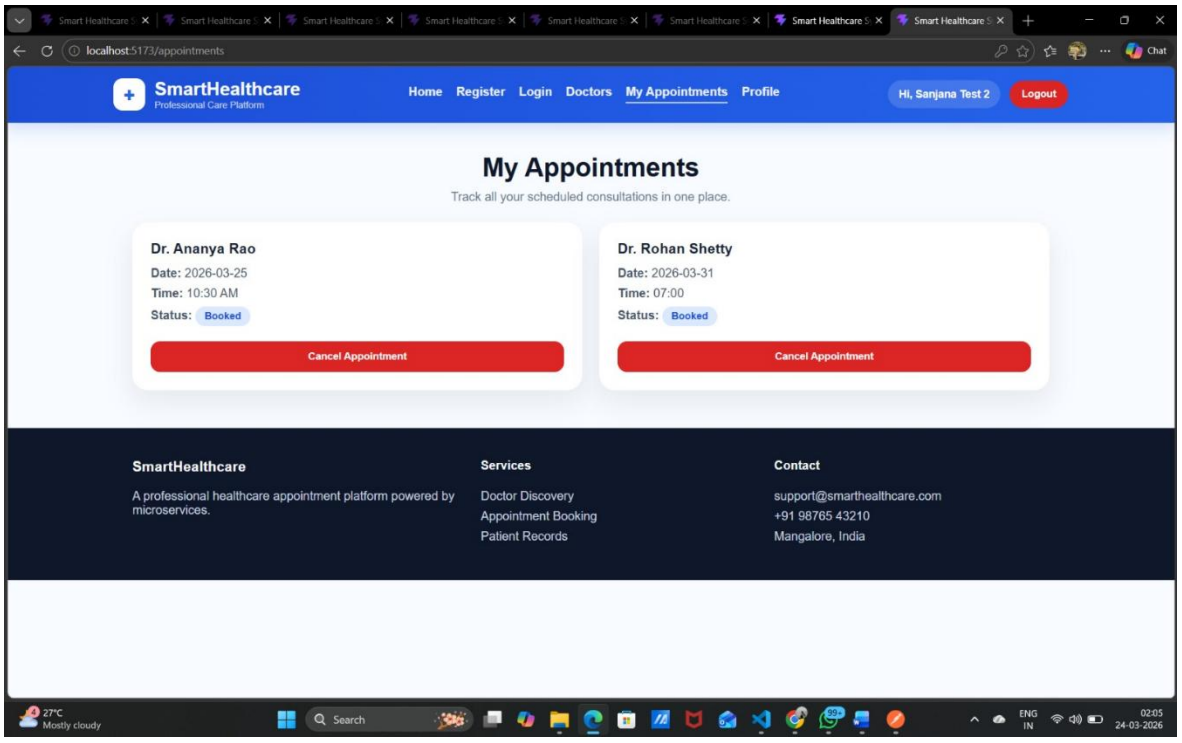

Fig 9.5 Appointments

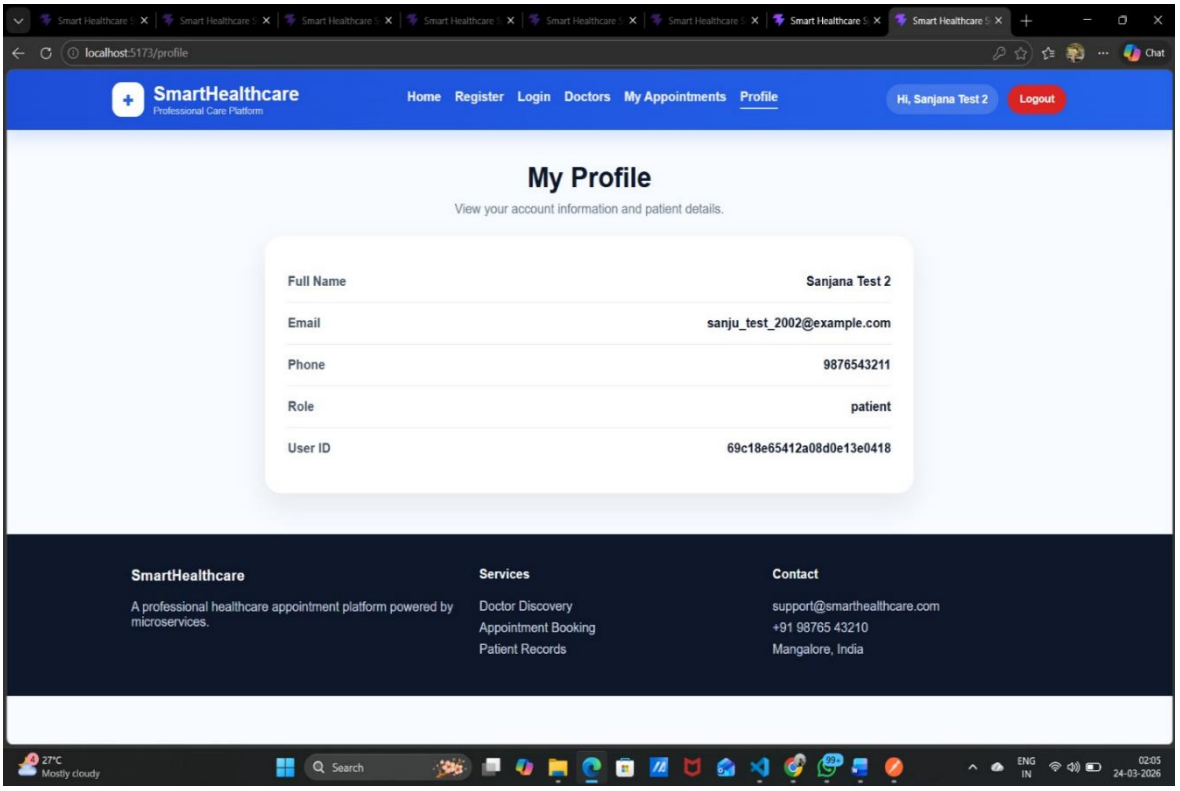

Fig 9.6 My profile

Chapter-10

ADVANTAGES & LIMITATIONS

10.1 Advantages

1. Easy Appointment Booking: Patients can register, log in, view doctors, and book appointments online, eliminating the need for physical visits.
2. Time-Saving: Reduces waiting time for patients.
3. Reduced Administrative Work: Automates appointment management and minimizes the workload for hospital staff.
4. Scalable Microservices Architecture: Each service (User, Doctor/Appointment, Notification) operates independently, allowing easy updates and maintenance without affecting the entire system.
5. Efficient Data Management with MongoDB: Secure storage of patient, doctor, and appointment information with quick retrieval and flexibility.
6. API Gateway for Smooth Communication: Ensures seamless interaction between frontend and backend services.
7. Real-Time Notifications: Sends appointment confirmations and reminders, reducing missed appointments.
8. User-Friendly Interface: Provides an intuitive interface for patients to navigate and manage appointments.
9. Accurate Record-Keeping: Maintains precise records of users, doctors, and appointments for better hospital management.
10. Improved Patient Convenience: Enhances the overall patient experience and efficiency of healthcare services.

10.2 Limitations

1. Internet Dependency: The system requires an active internet connection to function.
2. Limited to Registered Doctors: Only registered doctors are available for appointment booking.
3. No Emergency Support: Does not handle emergency consultations or walk-ins.
4. Notification Dependency: Appointment reminders depend on the accuracy of email or phone number provided by the user.
5. User Training Required: Some patients may require guidance to navigate the online system.
6. High System Load: Extremely high traffic may require additional server resources to maintain performance.

Chapter-11

FUTURE ENHANCEMENT

The Smart Healthcare Appointment System can be further improved with the following enhancements:

1. **Mobile Application Support:** Develop a mobile app for Android and iOS to allow patients to access the system on smartphones.
2. **Video Consultations:** Integrate a video conferencing feature for online doctor consultations.
3. **Payment Gateway Integration:** Enable online payments for consultations, appointments, or hospital services.
4. **AI-Based Doctor Suggestions:** Use artificial intelligence to recommend doctors based on patient history, symptoms, or specialization requirements.
5. **SMS and Push Notifications:** In addition to email, send appointment reminders via SMS and push notifications for better reliability.
6. **Hospital Dashboard for Admins and Doctors:** Provide a dashboard for doctors and hospital staff to manage appointments, view patient details, and generate reports.
7. **Emergency Appointment Handling:** Include features for urgent appointments or walk-ins to improve service in critical situations.
8. **User Reviews and Ratings:** Allow patients to rate doctors and provide feedback to improve service quality.
9. **Advanced Security Features:** Implement multi-factor authentication and encryption for better data protection.
10. **Analytics and Reporting:** Include reporting tools to analyze appointment trends, doctor performance, and patient statistics for hospital management.

Chapter-12

CONCLUSION

The Smart Healthcare Appointment System demonstrates how modern technologies can be used to improve healthcare service management. By leveraging a **microservices-based architecture**, the system divides functionality into independent services, such as User Service, Doctor/Appointment Service, Notification Service, and API Gateway. This design improves scalability, maintainability, and fault tolerance while ensuring that each module operates efficiently and independently.

The use of **React** for the frontend provides a responsive and user-friendly interface, allowing patients to easily register, log in, view doctors, and book appointments online. **MongoDB** is used as the database, providing secure, flexible, and scalable storage for patient, doctor, and appointment data. The integration of a Notification Service ensures that patients receive timely confirmations and reminders for their appointments, reducing missed consultations.

Overall, the system reduces manual workload, minimizes waiting time, and streamlines hospital management. It enhances patient convenience and provides a structured and organized method for handling appointments. The combination of modern web technologies, microservices architecture, and secure data management makes the system reliable, efficient, and adaptable for future enhancements.

In conclusion, the Smart Healthcare Appointment System provides a **comprehensive and scalable solution** for managing healthcare appointments, improving both patient experience and hospital workflow, while laying the foundation for further technological improvements in healthcare services.

Chapter-13

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- Microservices Architecture Guide: https://microservices.io/
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